5 **(a)** $E = \frac{V}{d} = \frac{45}{0.10} = 450\ \text{V m}^{-1}$ or $F_E = qE = \left(1.60 \times 10^{-19}\right) \times 450 = 7.2 \times 10^{-17}\ \text{N}$

$F_E = m\,a$

$e\,E = m\,a$

$(1.60 \times 10^{-19}) \times 450 = (9.11 \times 10^{-31}) \times a$

$a = 7.903 \times 10^{13}\ \text{m s}^{-2}$ or $7.9 \times 10^{13}\ \text{m s}^{-2}$

(b) $v_y = u_y + a_y\,t$

$0 = (6.8 \times 10^6 \times \sin 30°) + (-7.9 \times 10^{13})\,t$

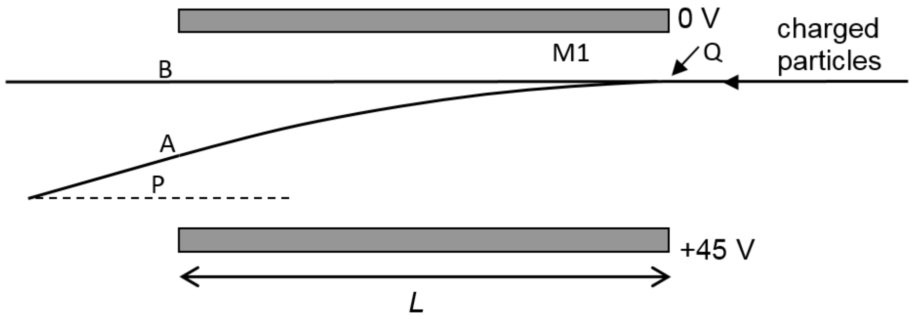
$t = 4.304 \times 10^{-8}\ \text{s}$

$s_x = u_x\,t + ½\,a_x\,t^{\,2}$

$L = (6.8 \times 10^6 \times \cos 30°) \times (4.304 \times 10^{-8}) + 0$

$L = 0.253$ m or 0.25 m

(c)



(i) Electrons from Q will have a smaller vertical displacement than previously as its horizontal speed now is higher. Path A exits above point P.

(ii) Mass of protons is 1800 times more than electrons. Vertical displacement of protons will be 1800 times less than that of electrons. Path B is almost straight and the upward vertical displacement is much less than Path A.

(iii) **I.** Path A and path B are deflected in opposite directions: due to opposite charges, the forces acting on electrons and protons are in opposite directions.

II. Path A shows a much greater deflection than Path B (Path B is nearly straight): Electrons and protons experience the same force. Due to the much smaller mass of electrons, they experience a much larger acceleration than protons and hence a larger vertical displacement.

6(a) $^{1}_{0}n + ^{235}_{92}U \rightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3\,^{1}_{0}n$